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Giancarlo Montes

# CYB102 | Intermediate Cybersecurity

Intermediate Cybersecurity Fall 2024 (@ Section 1 | Tuesdays 5PM - 7PM PDT)

Personal Member ID#: **113724**

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# Lab #6: Cat-a-logging Incidents

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## 🚩 MACHINE ACCESS CHECK 🚩

To proceed with this unit, please ensure you have access to a **cyb102 Ubuntu VM** as detailed in the [🔗 VM Access Guide](#).

We recommend you begin with **Azure Labs**.

✅ All options in the [🔗 VM Access Guide](#) will work for this unit.

## Overview

Catalyst is a powerful incident response management platform that provides a centralized hub for organizing, tracking, and collaborating on security incidents. It offers various features to support the incident response process, including incident creation, task management, observables tracking, and collaboration tools.

In this lab, we will be using Catalyst to manage and respond to security incidents. We have provided three incident response forms that document different security incidents: malware, brute force password attempts, and phishing emails. Your task is to use Catalyst to create incidents, input relevant information, and practice managing these incidents as if you were part of a real incident response team.

Generally in any incident response policy we'd have discussions about what systems we want to protect and how we are going to protect them. In Unit 5 we discussed Splunk which is a great way to detect when an incident is happening. Now we're going to discuss how to respond and how we'd recover when an incident would occur.



## Goals

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By the end of this lab, you will be able to...

- Create incidents in Catalyst for three different types of cyberattacks (malware, brute force, and phishing).
- Input and analyze observables (IoCs) for each incident based on provided incident response documents.
- Create tasks for documenting the incident timeline and summarizing key findings.
- Manage incident response workflows in Catalyst.

## Resources

Downloads:

- [ir\\_report\\_phishing.md](#)
- [ir\\_report\\_malware.md](#)
- [ir\\_report\\_brute\\_force.md](#)

Resources:

1. Catalyst User Guide: A comprehensive guide to help you understand Catalyst's features, installation, and usage.
2. Catalyst GitHub Repository: Access to the source code, issue tracking, and community contributions for Catalyst.
3. Introduction to Incident Response: A beginner-friendly resource to help you understand the basics of incident response.
4. NIST Computer Security Incident Handling Guide: A detailed guide on incident handling and response from the National Institute of Standards and Technology.
5. Cybersecurity Incident Response Planning: An overview of the steps required for effective incident response planning.

# Lab Instructions

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# Required Features

## Step 0: Set up

In the real-world, companies would have their own installations of Catalyst. Instead, we'll be **installing a local Catalyst server** onto our Ubuntu VM.

### To install Catalyst: (🌟 NEW)

On a terminal in your Ubuntu VM, run the following command to download [the script](#):

- `wget https://raw.githubusercontent.com/codepath/cybl02-file-storage/main/unit6_fix_sp24.sh`

Then make the script executable:

- `chmod +x unit6_fix_sp24.sh`

Finally, run the script:

- `sudo ./unit6_fix_sp24.sh`
- This will install Catalyst and set it up for you to use! It should take about 5 minutes to complete.

When the script is done, you should see a message that says `[CATALYST SETUP] Catalyst is running. Try connecting at https://catalyst.localhost` If you see that, you're all set to move on to the next step.

### To access Catalyst: (🌟 NEW)

Make sure you have a **desktop connection** to your Ubuntu VM.

From the VM's desktop, follow along with  [this video \(no sound\)](#) to get logged in and set up.

Use the following credentials:

- Username: admin

- Password: admin

**TIP** Need to  start /  stop Catalyst? Click here for commands!

 Those credentials seem... unsafe. Why do you think it's okay to use them here?

 **Checkpoint 0:** You should now be logged into Catalyst! Click around a bit and familiarize yourself with the Catalyst user interface and navigation.

## Step 1: We're on the Case!

Before we get into it, take a moment to familiarize yourself with the available features in Catalyst:

Alerts

Open Tasks

Forensic Investigations

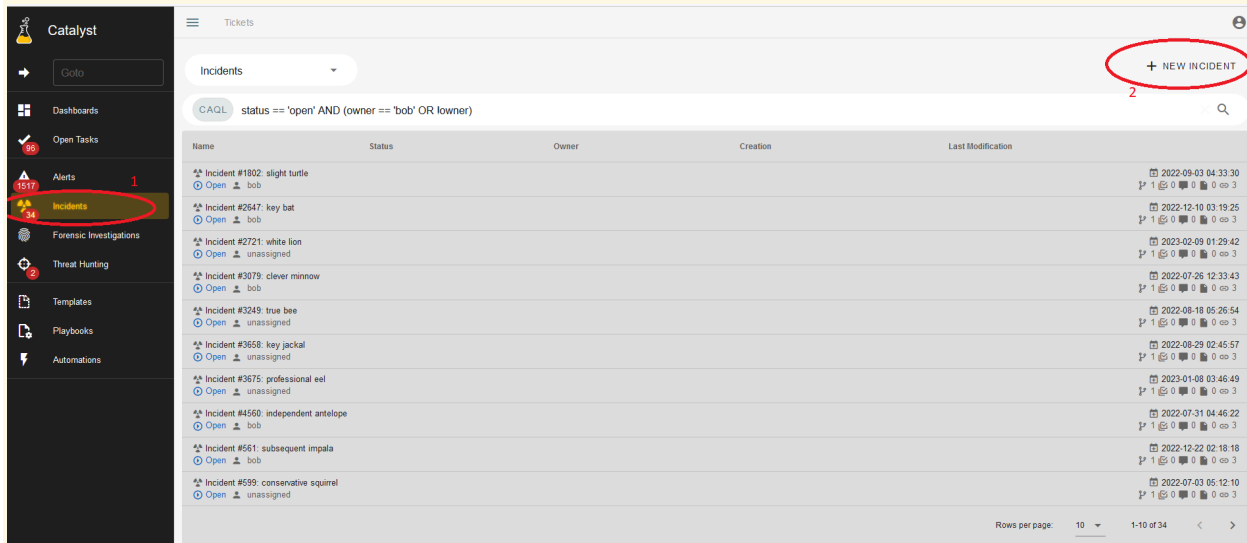
Threat Hunting

Explore the different sections and tabs within the case to understand how information is organized and displayed. Take note of any additional features or functionalities in Catalyst that might be useful for incident response and investigation.

### 1.1: Creating an incident

Alright, create our first incident!

First, we'll need to access our phishing incident response report [ir\\_report\\_phishing.md](#), then bring that template into Catalyst.



In Catalyst, navigate to the incident management section by clicking Incidents.

Click on the "New Incident" button to create a new incident.

Using the incident report ( [ir\\_report\\_phishing.md](#)), try filling in the details for the new incident.

Once you've finished, click each field below and check your work:

Title

Playbooks

Template

Severity

TLP

- 
- 
- 
- 

Description

- 

Once you've filled in all the required fields:

Click the "Create" button to create the incident.

Stuck? Click here to view a  screenshot!

Incidents

+ NEW INCIDENT

CAQL

status == 'open' AND (owner == 'bob' OR

Incident #1802: slight turtle

Open

bob

2022-09-03 04:33:30

Incident #2647: key bat

Open

bob

2022-12-10 03:19:25

Incident #2721: white lion

Open

unassigned

2023-02-09 01:29:42

Incident #3079: clever minnow

Open

bob

2022-07-26 12:33:43

Incident #3249: true bee

Open

unassigned

2022-08-18 05:26:54

Incident #3658: key jackal

Open

unassigned

2022-08-29 02:45:57

Incident #3675: professional eel

Open

unassigned

2023-01-08 03:46:49

Incident #4560: independent antelope

Open

bob

2022-07-31 04:46:22

Incident #561: subsequent impala

Open

bob

2022-12-22 02:18:18

Incident #599: conservative squirrel

Open

unassigned

2022-07-03 05:12:10

Rows per page: 10

1-10 of 34

NEW INCIDENT

IR-2023-001

Phishing

Default

High

Green

Description

A phishing email was received by the user, Jane Smith, containing a malicious attachment. Upon opening the attachment, malware was executed on the affected system (PC-001). The malware was identified as a known ransomware variant.

CREATE

 **Checkpoint 1:** You've created an incident in Catalyst!

## 1.2: Breaking it down

***Okay, we've created an incident in Catalyst! Why? What does this do for us?***

Well, but using Catalyst, we can organize each incident that happens in our environment and assign out responsibilities as we figure out what is going on. But that only works if everyone is on the same page.

***So, what exactly is an incident in Catalyst?***

An incident in Catalyst represents a specific security event or issue that needs to be addressed. It serves as a container to capture and manage all the relevant information, actions, and

analysis related to that incident. Within each incident, there are various elements, such as tasks, observables, and TTPs (Tactics, Techniques, and Procedures).

Let's take a closer look at each of these elements:

1. **Tasks:** These are individual action items or steps that need to be taken by a cybersecurity analyst or incident response team member to respond to the incident effectively. Tasks help break down the incident response process into manageable chunks and assign responsibility to specific team members.
2. **Observables:** Observables are pieces of data or evidence related to the incident, such as IP addresses, file hashes, or email addresses. Analysts use observables to identify patterns, relationships, and trends that may help them understand the scope of the incident and how to mitigate it.
3. **TTPs:** Tactics, Techniques, and Procedures represent the behavior patterns and methodologies used by threat actors during an attack. Understanding the TTPs associated with a specific incident can provide valuable insight into the attacker's motives, capabilities, and potential future actions.

In the rest of this lab, we will experiment with these different incident categories. To get started, let's add relevant *observables* to our incident.

## Step 2: The Investigation Begins

### 2.1: Add a simple Artifact

Navigate to the "Open Tasks" section.

Here, find the incident you created earlier, it should be on the last page.

On the right hand side you'll notice that there are a few things you can add including, references, Artifacts, Related Tickets, and files.

We're going to be adding in the Artifacts on the right hand side. These are going to be things like IP addresses, MD5 Hashes, and Domain Names.

Try to add the Malicious IP from the Phishing report as an Artifact. Then press the  button on `hash.sha1` to generate its' hash.

 Click here to check your work

## Artifact

45[.]142[.]166[.]228

Status:  Unknown · Kind:  · Type: ip

hash.sha1



hash: ab877b1d37b487938cbf8ec0bff680c4dd46b4d0

vt.hash



 CANCEL

Add the Malware Hash to your Artifacts as well.

 **Checkpoint 2:** You have investigated the phishing attack and documented your findings in Catalyst.

**Q:** Why does the IP address have all the [.] in it?

**Q:** Are "known as malicious" IPs reliable indicators?

## **2.2: Investigate IoCs using external resources**

Now that we have added observables to our incident, let's use some outside tools to get more context and information about each observable.

**IoCs** (Indicators of Compromise) are pieces of data that can be used to identify malicious activity. These can be things like IP addresses, MD5 Hashes, and Domain Names. We can use these to search for other instances of the same IoC, and see if it has been reported as malicious before, using external resources like [VirusTotal](#) and [AbuseIPDB](#).

### **Virus Total:**

Open a new browser tab and navigate to [VirusTotal](#).

In VirusTotal, locate the search bar at the top of the page.

For file hashes, copy the hash from your observable in Catalyst and paste it into the search bar in VirusTotal

Press Enter to search for the hash.

For IP addresses, copy the IP address from your observable in Catalyst and paste it into the search bar in VirusTotal. Press Enter to search for the IP address.

Investigate the information provided by VirusTotal for each observable and make note of any relevant details, such as the file's reputation or the IP address's associations with malicious activities.

### **AbuseIPDB:**

Open another new browser tab and navigate to [AbuseIPDB](#).

In AbuseIPDB, locate the search bar near the top of the page.

Copy the IP address from your observable in Catalyst and paste it into the search bar in AbuseIPDB. Press Enter to search for the IP address.

Investigate the information provided by AbuseIPDB for the IP address and make note of any relevant details, such as the IP address's abuse history or the types of reported activities.

Let's add our findings to Catalyst.

In Catalyst, add notes to each observable with the information gathered from VirusTotal and AbuseIPDB in the log.

This additional context can be valuable for other analysts working on the incident and may help inform decisions on further actions to take in response to the incident.

Click on each artifact and mark it as Malicious, Safe, or Unknown.  
Click on each Artifact its' type as an IP address, MD5 Hash, etc.

 **Checkpoint 3:** At this point, you've documented IoCs for your incident in Catalyst and investigated them using external resources. Nice!

### Step 3: Closing the Case

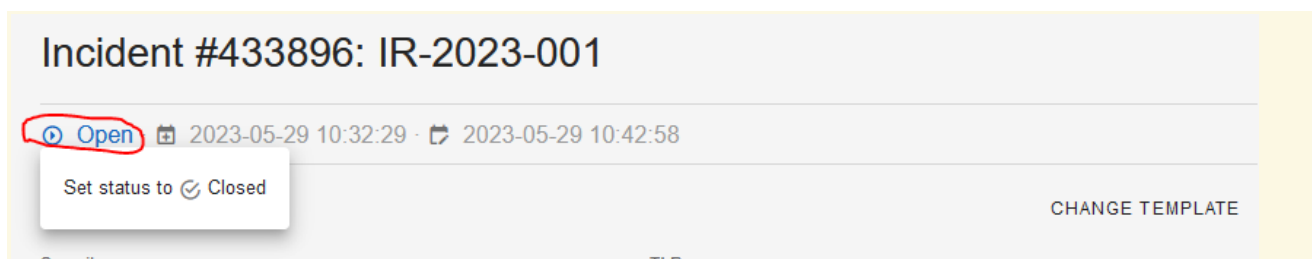
Alright, at this point we've determined what happened, what's happening now, and it seems like there's nothing further that needs to be done for this incident!

Document the resolution details in the "Description" section.

- This might include what actions were taken to resolve the incident, what changes were made to systems or processes, and what steps will be taken to prevent similar incidents in the future.

Click on the blue text that says "Open", and set the current ticket to Set Status to "Closed".

- Note: Do not click Close in the top right -- that will simply exit the ticket without changing its status!



In Catalyst, closing an incident means marking it as **resolved** and **completing the incident response process**. This signifies that all tasks, observables, and TTPs have been properly documented, and the incident has been thoroughly investigated.

 **Checkpoint 4:** Great work! At this point, your phishing incident should be fully resolved in Catalyst. 

## Step 4: Lessons Learned

This next step is a little different than the previous ones, and might feel like it's something out of an English class! Trust us, we're only showing you things real-world cybersecurity professionals have to do -- and reporting your findings is a key skill.

#### 4.1: What are Lessons Learned?

In cybersecurity, once an incident has been responded to and eradicated, it's crucial not to just move on to the next thing. A critical component of incident response is the **"Lessons Learned"** phase, sometimes also referred to as *"Post-Incident Activity"* or *"After-Action Review"*. This step involves analyzing the incident and the response to identify what was done well, what could have been done better, and what changes can be made to prevent similar incidents in the future.

By documenting lessons learned, an organization can continuously improve its incident response capabilities. The goal is to make each response better than the last.

#### 4.2: Classifications

In addition, every incident that's handled by an incident response team is classified, usually in one of four ways:

- **True Positive:** The incident is confirmed as a genuine security event that requires remediation. These are actual incidents that warrant a response.
- **False Positive:** The incident is a false alarm, and no further action is needed. These can be caused by errors in detection tools or misinterpretations of benign activities.
- **Indeterminate:** It's unclear whether the incident is a true or false positive, and additional investigation may be required. This classification is often used when there's not enough information to make a definitive decision.
- **Duplicate:** The incident is a duplicate of another incident that has already been investigated. Duplicates can occur if the same incident triggers multiple alerts in different systems.

#### 4.3: Lessons Learned for our Phishing Incident

Okay, that was a lot of reading... let's try writing our own Lessons Learned and classification, based on the phishing incident we just investigated.

Navigate back to each incident detail page in Catalyst. Reflect on the information collected, the steps taken, and the results of your investigation.  
In a new document, create a "Lessons Learned" section for the incident.

Here's an outline you can use:

**Incident Summary:** What happened? What was the impact of the incident?

**Tactics, Techniques, and Procedures (TTPs):** What did you learn from this incident about the TTPs used by attackers? How did the attacker(s) gain access, and what did they do once they were inside the network?

**Response Effectiveness:** What steps were effective in responding to the incident? Did you quickly identify the issue, contain the damage, and eradicate the threat?

**Improvement Areas:** What steps could have been handled better? Were there delays in detection, gaps in communication, or issues with the tools you used?

**Future Changes:** What changes would you suggest for future responses based on your experience with this incident? This could include changes to tools, policies, procedures, or training.

After reflecting on the incident, determine its classification. Write down your chosen classification and explain your reasoning.

🎉 Congratulations, you've completed the required portion of the lab! If you have more time or want to practice further, check out the stretch features below.

## Stretch Features

### Step 5: The Other Datasets

At the beginning of this lab, we gave you three incident reports... but we only had you investigate one of them! If you're feeling up to it, try investigating the other two incidents as well.

- [ir\\_report\\_malware.md](#)
- [ir\\_report\\_brute\\_force.md](#)

Remember to follow each of the steps we completed above:

- Create a new incident in Catalyst for each incident report.
- Add the incident details to the "Description" section.
- Add the observables as Artifacts.
- Investigate the observables using external resources.
- Add notes to each observable with the information gathered from external resources.
- Mark each artifact as Malicious, Safe, or Unknown.
- Mark each artifact with its' type as an IP address, MD5 Hash, etc.
- Close the incident in Catalyst.
- Document the Lessons Learned for each incident.
- Determine the classification for each incident.

